

Supplementary Appendix

“Decentralized Stochastic Energy-Reserve Co-Optimization and Pricing in Coupled Electricity-Gas Networks”

This appendix accompanies the main manuscript and collects the detailed discussions that supplement the main text: Appendix A discusses the accuracy of the GSF linearization and compares it with Second-Order Cone Relaxation (SOCR)-based formulations (supplementing Section II-B of the main paper); Appendix B expands the future research directions briefly mentioned in Section VII of the main paper; Appendix C discusses the key assumptions underlying the pricing framework (supplementing Section IV-C of the main paper); and Appendix D elaborates the practical implications for system operators (supplementing Section VI of the main paper). The equations restated at the end of this appendix carry the same numbers as in the main paper. All added content is highlighted in blue.

APPENDIX A

ACCURACY OF THE GSF LINEARIZATION AND COMPARISON WITH SOCP-BASED FORMULATIONS

The proposed GSF-based linearization replaces the nonlinear Weymouth equation of the main paper, $f_l = K_l \sqrt{\pi_m^2 - \pi_n^2}$, with a first-order Taylor expansion around a predicted baseline operating point $(\pi_{m,0}, \pi_{n,0})$. The approximation error of this linearization is governed by the deviation of actual nodal pressures from the reference values. In high-pressure gas transmission networks—where pressure deviations between adjacent nodes are typically within 5%–10% of their nominal values under normal operation—the truncation error of the first-order expansion is of order $\mathcal{O}(\delta^2)$, where $\delta = \max_{m,n} |\pi_m - \pi_{m,0}| / \pi_{m,0}$ is the maximum relative pressure deviation. For the 20-node network studied in Section VI of the main paper, the reference pressures $(\pi_{m,0}, \pi_{n,0})$ are obtained from a prior load-flow solution using forecasted base-case demands. This carefully calibrated baseline yields mild relative pressure deviations across all nodes. Following the $\mathcal{O}(\delta^2)$ error scaling property of first-order Taylor linearization, the resulting flow mismatch between the GSF linear approximation and the original nonlinear Weymouth model remains tightly bounded.

It is instructive to contrast this approach with the Second-Order Cone Relaxation (SOCR)-based gas flow formulation, which is the predominant convex alternative [1], [2]. SOCR replaces the nonconvex Weymouth equality with an inequality constraint $f_l^2 \leq K_l^2(\pi_m^2 - \pi_n^2)$ and requires the relaxation to be *exact* (i.e., the inequality binds at optimality) for the resulting dual variables to constitute valid marginal prices. However, exactness of SOCR is not guaranteed a priori: it depends on the specific objective function, network topology, and load conditions, and counterexamples exist where the relaxation gap leads to physically infeasible flow solutions and distorted price signals [2]. In contrast, the proposed GSF approach enforces strict equality of the linearized flow constraints by construction, thereby guaranteeing that the KKT conditions yield valid, duality-gap-free dual multipliers regardless of the

operating point. This property is critical for the financial settlement and the theoretical proofs of market properties established in Section IV of the main paper. The trade-off, therefore, is between the controllable approximation error of GSF linearization (mitigated by selecting an informed baseline operating point, e.g., from a previous market run or a load-flow initialization) and the contingent but unpredictable relaxation gap of SOCR.

Regarding the penalty constant K in (6) and (8) of the main paper, a value of $K = 10^4$ – 10^5 times the largest pipeline coefficient K_l is recommended. Within this range, the compressor ratio enforcement residual $|\rho_c \pi_m - \pi_n|$ is driven below 10^{-5} p.u., making the compressibility violation numerically negligible for all practical purposes. Sensitivity tests on the test system confirm that the clearing results and dual multipliers are insensitive to K in this range, as the penalty term serves solely to enforce the compressibility equality (5) without affecting the objective function value. Excessively large values beyond 10^6 should be avoided to prevent numerical ill-conditioning.

APPENDIX B

FUTURE RESEARCH DIRECTIONS

This appendix expands the brief research outlook given in Section VII of the main paper. Future work will extend the proposed framework along several directions: (i) incorporating multi-period gas dynamics to capture the temporal flexibility of pipeline linepack and its implications for inter-temporal price formation and storage valuation; (ii) integrating hydrogen blending and power-to-gas facilities into the coordinated pricing framework, addressing the additional nonconvexities introduced by electrolyzer efficiency curves and hydrogen fraction constraints; (iii) coupling carbon emission markets with the electricity-gas co-clearing mechanism to derive joint energy-reserve-carbon price signals and to quantify the abatement cost of decarbonization targets across networked energy sectors; and (iv) extending the analysis to strategic multi-agent settings where GFU operators or gas suppliers exercise market power, enabling the study of market design interventions to mitigate cross-network market power.

APPENDIX C

DISCUSSION ON THE KEY ASSUMPTIONS UNDERLYING THE PRICING FRAMEWORK

The proposed pricing framework and the associated market property proofs rest on several modeling and theoretical assumptions that warrant explicit discussion to ensure logical transparency and to preempt potential reviewer concerns.

1) *Zero Lower Generation Limits* ($\underline{P}_i = 0, \underline{Q}_j^S = 0$). The individual rationality proof in Property 2 of the main paper assumes zero lower bounds for generators and gas sources. This assumption aligns with standard short-term market practice where minimum stable output (technical minimum) is typically handled through unit commitment decisions made prior to the economic dispatch stage, rather than being enforced as a binding constraint during the pricing phase. In a day-ahead co-clearing context, generation units that cannot operate

profitably at prevailing prices would submit decommitment bids or self-schedule to zero. If the assumption is relaxed and strictly positive lower bounds are present, the expected profit expression in (56) of the main paper acquires an additional term $-\nu_i^{ED} \underline{P}_i$, whose sign depends on the shadow price ν_i^{ED} of the lower bound constraint. In practice, ν_i^{ED} is zero for units dispatched strictly above their minimum, and non-negative when the lower bound binds—the latter case implying the unit is marginal and earns zero profit, which remains consistent with individual rationality.

2) *Convexity and Strong Duality.* The GSF-based linearization renders the clearing model a linear program (LP). For LPs, strong duality holds unconditionally provided a feasible solution exists; no constraint qualification or exactness condition needs to be verified independently. This is a key advantage over SOCR-based pricing frameworks, where strong duality is contingent on the exactness of the conic relaxation at the optimum. The analytical price decompositions in (39)–(44) of the main paper and the proofs of Properties 1–3 therefore hold without additional qualification, provided the clearing problem is feasible.

3) *Feasibility of All Scenarios.* The stochastic programming formulation requires that every scenario $k \in \mathcal{K}$ admits a feasible recourse solution. In practice, this is ensured by the inclusion of load shedding variables $p_{i,k}^{\text{shed}}, q_{j,k}^{\text{shed}}$ with sufficiently high VOLL penalties V^E, V^G , which serve as slack variables that guarantee feasibility under any realization of uncertain demand. The VOLL penalties are set to sufficiently high standard regulatory values for electricity and gas, ensuring that load shedding is activated only under extreme conditions that would otherwise render the problem infeasible. The zero load shedding result observed in the case study (Section VI of the main paper) confirms that the procured reserve capacities are sufficient to absorb the modeled uncertainty, validating the practical relevance of the feasibility assumption.

4) *GSF Matrix Dependence on Operating Point.* The GSF matrix H^G and the scenario-specific variants H_k^G are constructed around the predicted baseline pressure profile $(\pi_{m,0}, \pi_{n,0})$, which is obtained from a prior load-flow solution using forecasted demands. For the scenario-specific matrices H_k^G , the baseline pressures are recomputed using each scenario's demand realization to reflect the altered flow directions and pressure drops. The sensitivity of the GSF approximation to the choice of baseline operating point is mitigated by the fact that gas network topology and compressor ratios are invariant across scenarios; only the pressure magnitudes shift modestly in response to load variations. As discussed in Appendix A, the approximation error introduced by the GSF linearization stays at a low level for all test cases.

APPENDIX D

PRACTICAL IMPLICATIONS FOR SYSTEM OPERATORS

The proposed framework carries several actionable implications for the operation and design of coupled energy markets.

1) *For Electricity System Operators (ISO-E):* The decomposed LMP expression (39) of the main paper reveals that the scarcity component $\Xi_n^{E,\text{Res}}$ can account for up to 47% of

the nodal price at congested locations (see Table III of the main paper), meaning that nearly half the value of delivered energy at these buses stems from uncertainty hedging rather than from the energy commodity itself. This decomposition equips ISO-E with a quantitative tool to identify which buses face the most acute exposure to multi-energy demand risk, guiding siting decisions for flexible resources and transmission expansion. Furthermore, the gas-to-power congestion propagation mechanism (41) of the main paper provides ISO-E with early warning of emerging gas-side constraints before they materialize as power-side reliability events.

2) *For Gas System Operators (ISO-G):* The LMGP decomposition (44) of the main paper exposes the spatial distribution of pipeline congestion through its congestion component $(H^G)^T(\mu^{G+} - \mu^{G-})$ and the endogenous loss propagation factor $(I + EH^G)^{-T}$. The latter reveals that a marginal increase in demand at a downstream node raises the system-wide gas price not only by the energy cost but also by the incremental compressor fuel consumption induced along the delivery path—a physically intuitive effect that is quantitatively captured by the proposed pricing formula. ISO-G can use these spatial price differentials to prioritize pipeline reinforcement or storage investments at segments where congestion premia are persistently large.

3) *For Market Design and Regulation:* The formal proofs of revenue adequacy and individual rationality (Properties 1–2 of the main paper) establish that the coordinated mechanism is financially self-sustaining and incentive-compatible without requiring out-of-market uplift payments or make-whole settlements—an essential condition for regulatory approval. The ADMM implementation demonstrates that coordination can be achieved under existing institutional separation, exchanging only GFU gas consumption quantities between operators while preserving the confidentiality of network topologies and generator bids.

4) *For Cross-Network Risk Management:* The scenario-wise re-dispatch results (Table V of the main paper) confirm a one-to-one correspondence between procured reserves and actual recourse, validating that the stochastic formulation produces physically realizable hedging strategies rather than paper savings. The cross-network price propagation analysis (Fig. 3 of the main paper) demonstrates that gas-side congestion at a single node can amplify the LMP at a distant electrical bus by a factor of up to $9\times$ through the GFU conversion coefficient, providing both operators with a quantitative basis for joint contingency planning.

Restated Equations from the Main Paper

For the reader's convenience, the equations of the main paper referenced above are restated below with their original equation numbers.

$$\rho_c \pi_m - \pi_n = 0. \quad (5)$$

$$D_{l,l} = \begin{cases} K_l / \sqrt{\pi_{m,0}^2 - \pi_{n,0}^2}, & l \text{ as pipeline} \\ K, & l \text{ as compressor} \end{cases} \quad (6)$$

$$f_c = K(\rho_c^2 \pi_{m,0} \pi_m - \pi_{n,0} \pi_n) = K \rho_c \pi_{m,0} (\rho_c \pi_m - \pi_n). \quad (8)$$

$$\text{LMP}_n = \frac{\partial \mathcal{L}}{\partial p_n^D} = \lambda^E - \sum_{l \in \Omega_E^L} H_{l,n}^E (\mu_l^{E+} - \mu_l^{E-}) + \Xi_n^{E,\text{Res}} \quad (39)$$

$$\text{LMP}_{n(i)} = \phi_i \cdot \text{LMGP}_{m(i)} + \bar{\nu}_i^{EU} - \underline{\nu}_i^{ED} \quad (41)$$

$$\text{LMGP} = (I + EH^G)^{-T} [\lambda^G \cdot \mathbf{1} - (H^G)^T (\mu^{G+} - \mu^{G-})] + \Xi^{G,\text{Res}}. \quad (44)$$

$$\begin{aligned} \mathbb{E}[\pi_i] &= (\bar{\nu}_i^{EU} - \underline{\nu}_i^{ED}) p_i^{G*} + \bar{\nu}_i^{EU} r_i^{EU*} + \underline{\nu}_i^{ED} r_i^{ED*} \\ &= \bar{\nu}_i^{EU} (p_i^{G*} + r_i^{EU*}) - \underline{\nu}_i^{ED} (p_i^{G*} - r_i^{ED*}). \end{aligned} \quad (56)$$

REFERENCES

- [1] L. Dong *et al.*, “Distributed restoration for integrated electricity-gas-heating energy systems with an iterative loop scheme,” *Energy*, vol. 304, p. 132051, Sep. 2024.
- [2] H. Bai *et al.*, “A review of the application of locational marginal price theory in the new situation,” in *Proc. 3rd IEEE Conf. Energy Internet Energy Syst. Integr. (EI2)*, 2019, pp. 2126–2131.